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(54) **EXCHANGEABLE EXERCISE WEIGHTS**

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(57)

ABSTRACT

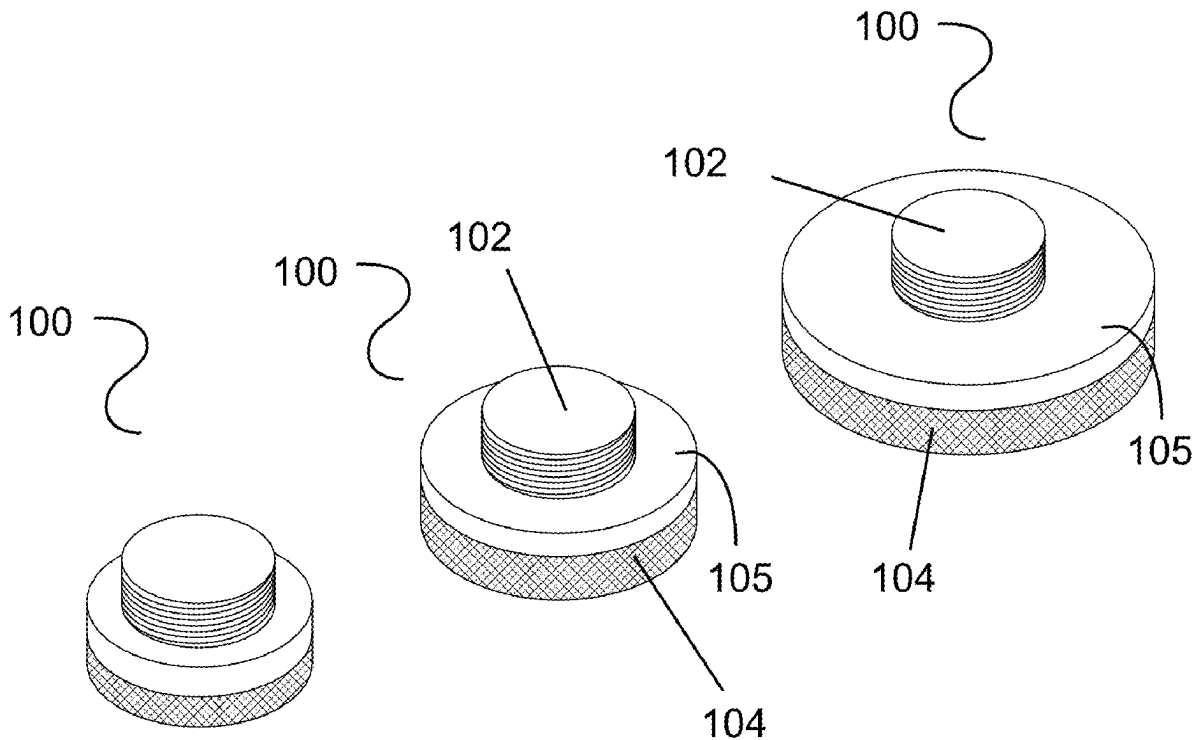
This application discloses exercise weights compatible with different types of handles and are removeable from handles for exchange. The exercise weights have connecting mechanisms to connect with corresponding connecting mechanism on handles to form exercise apparatuses. The exercise weights can connect with each other as well as the handles to increase weight load on the handles. Upon attaching to handles, exercise weights as disclosed herein are optionally hidden inside the handles, such that the exercise apparatuses overall have the appearance of common exercise apparatuses and are convenient for storage.

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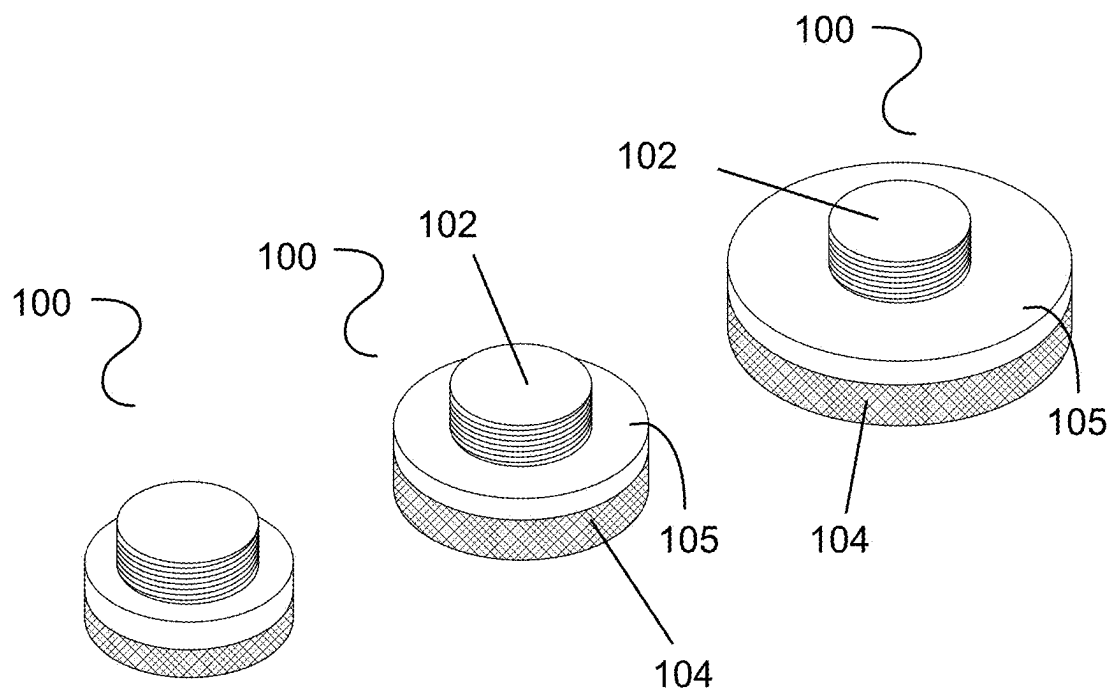


FIG. 1

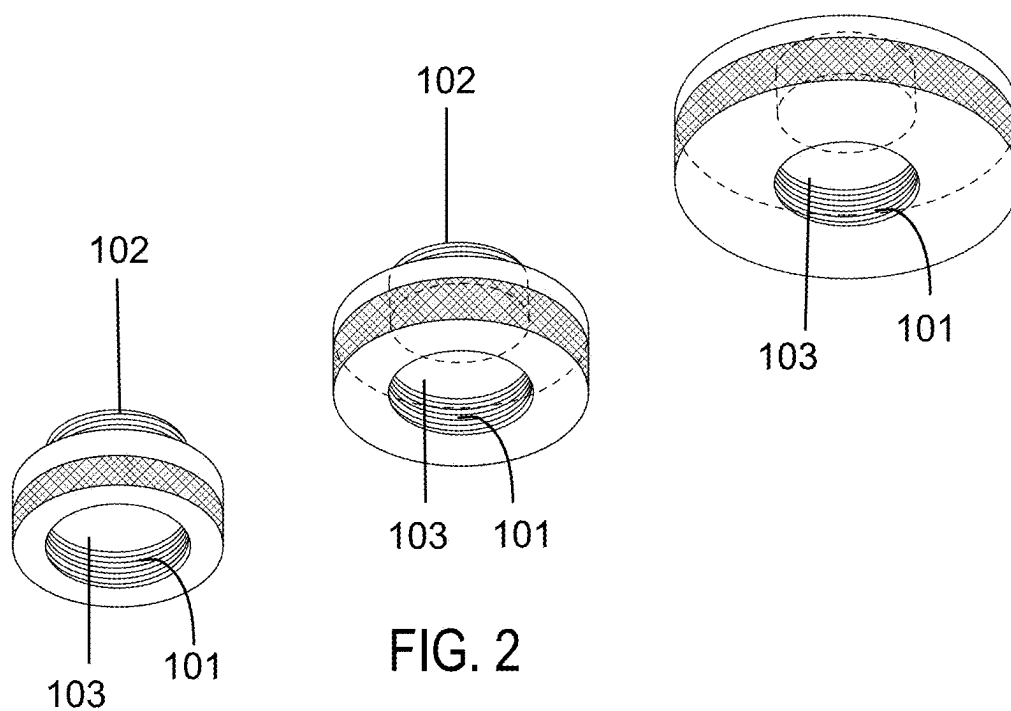
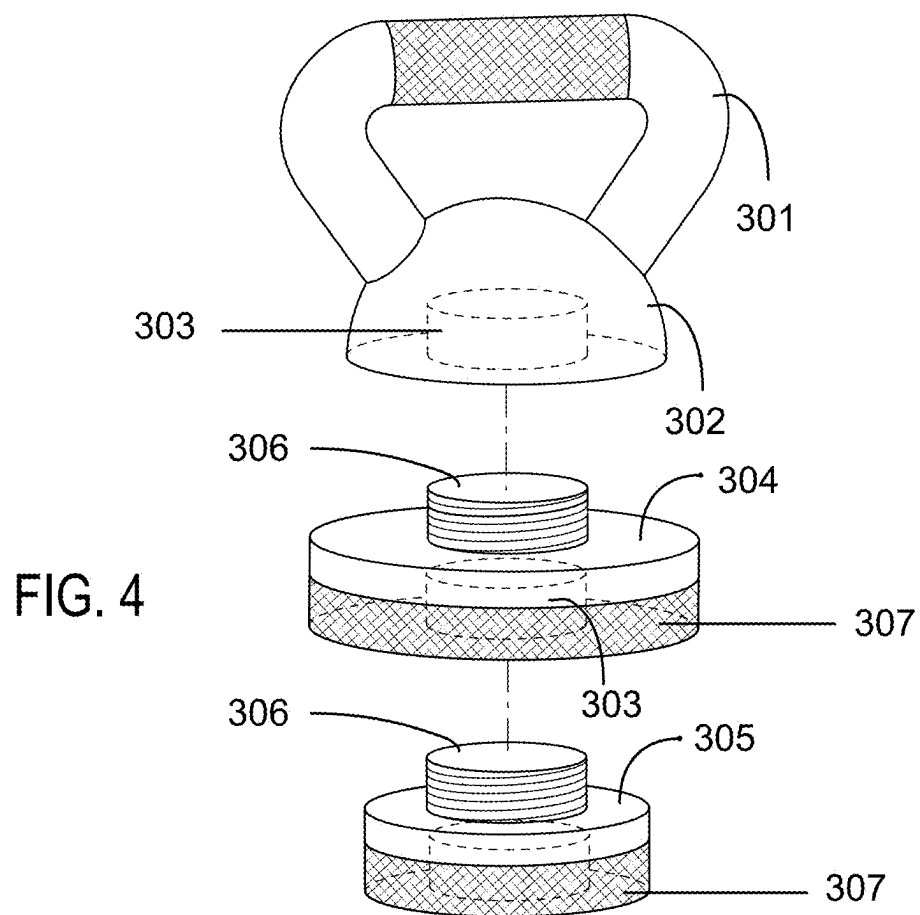
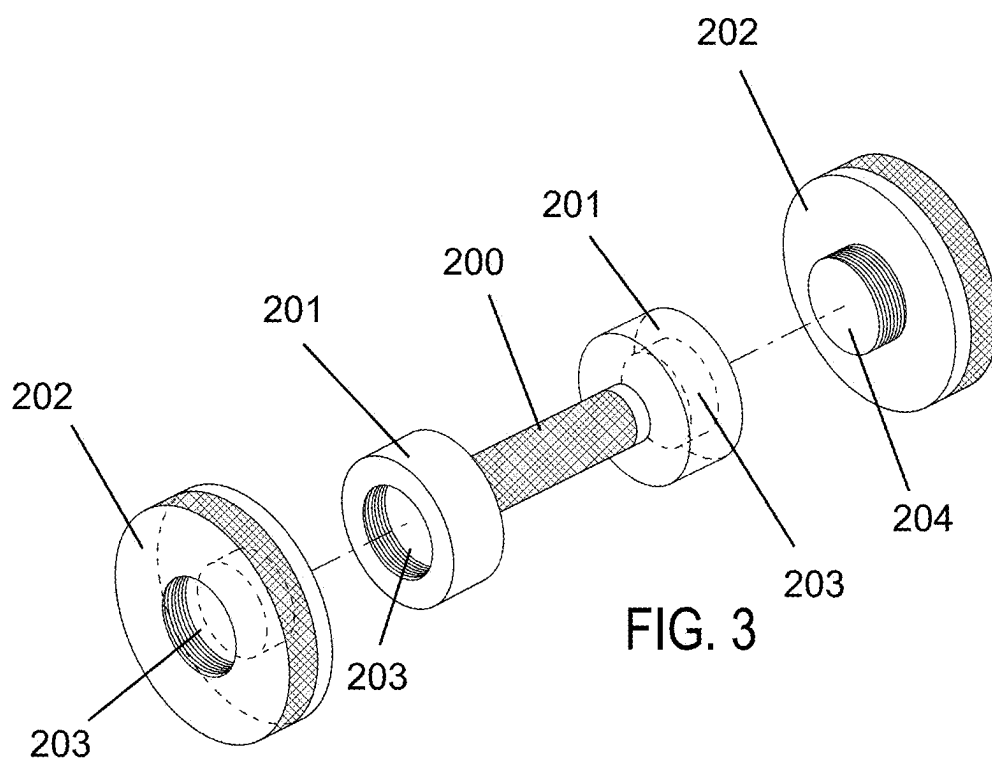
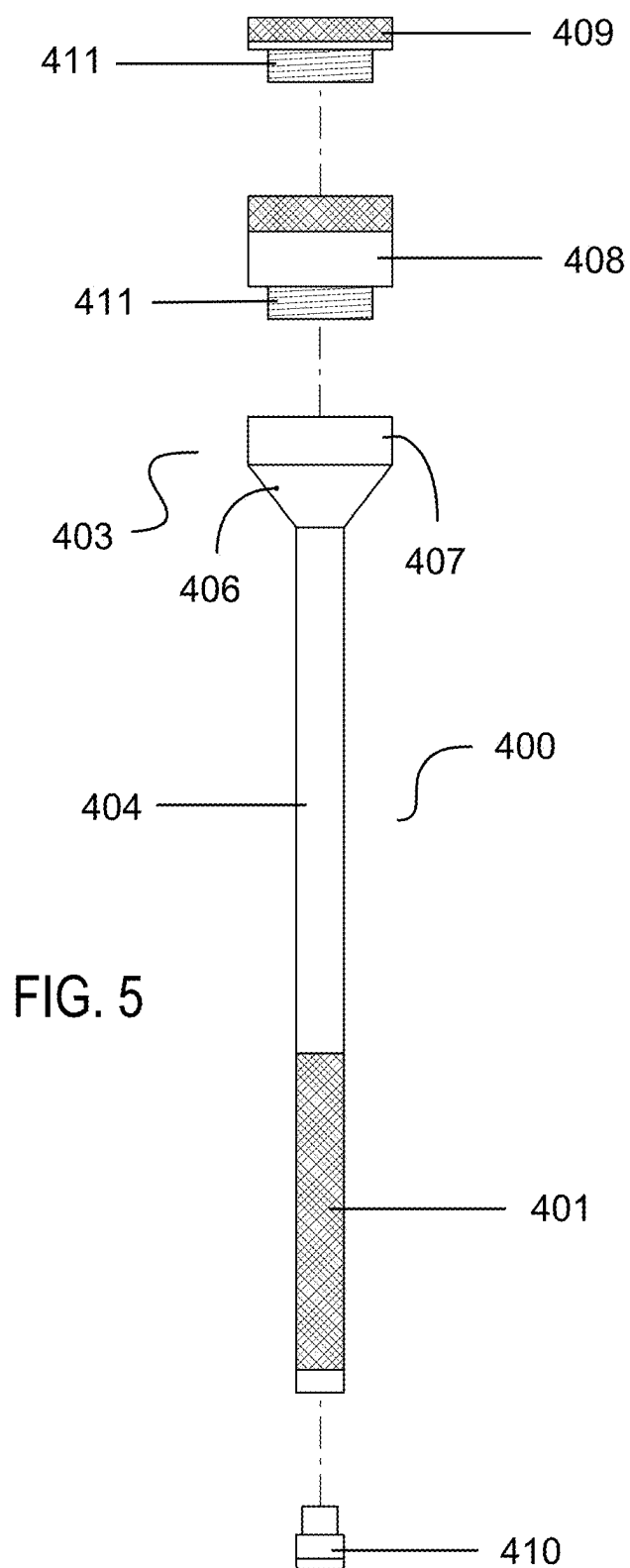


FIG. 2





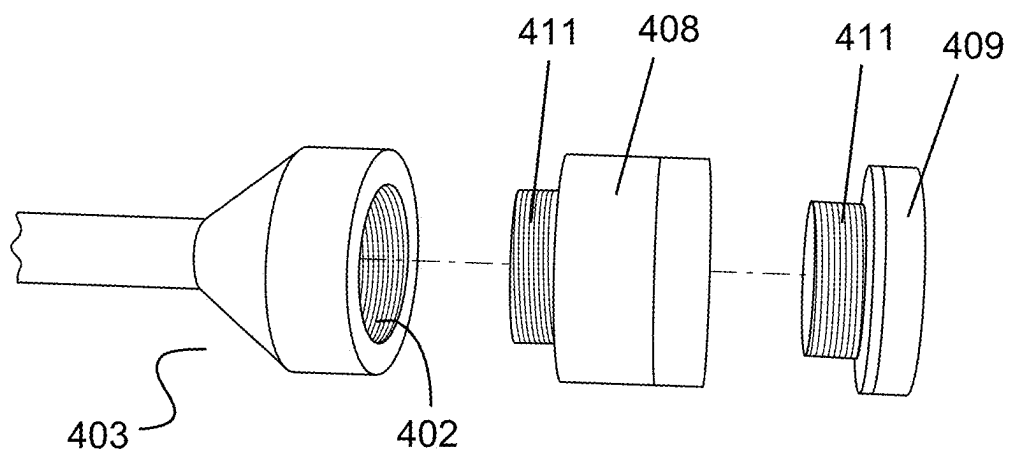


FIG. 6

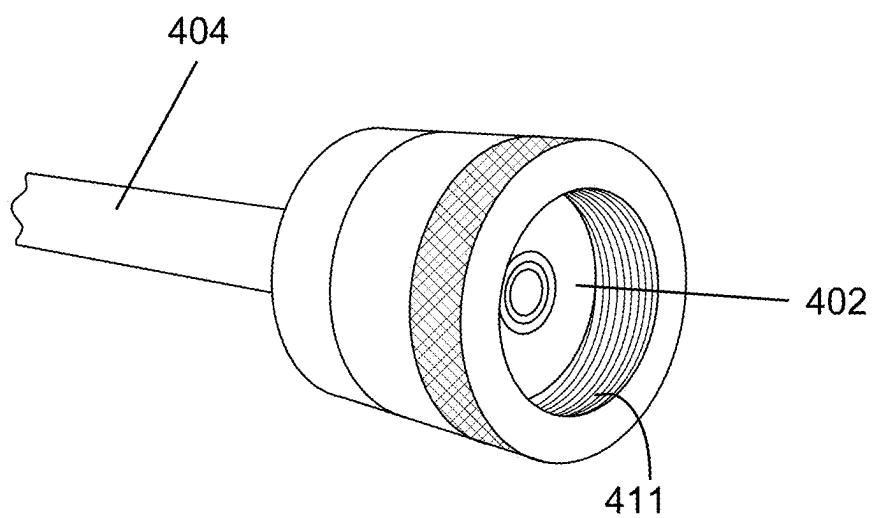


FIG. 7

EXCHANGEABLE EXERCISE WEIGHTS**BACKGROUND OF THE INVENTION****Field of the Invention**

[0001] This invention is in the field of exercise equipment. The invention disclosed herein concerns weights for use in exercising.

Description of the Related Technology

[0002] Lifting weights for exercise purposes is a part of exercise routines. Weights are designed for isolating groups of muscle during use. Typically, weights are designed for arm muscle groups, leg muscle groups, shoulder muscle groups, and back muscle groups.

[0003] Each person requires a different weight set due to their specific strength and weightlifting ability. It is common to see a weight rack of many different weights of the same type but of varying heaviness. For one person, a weight set of different weights, but the same type may be needed. For example, a person may need a dumbbell of 20 pounds for bicep exercise, but a dumbbell of 10 pounds for triceps exercise.

[0004] To reduce the number of pieces of weight and increase weight versatility, different methods have been used. Exercise equipment for weightlifting purposes is provided with a main part with an adapter to switch out weights. A typical example is a bar with two weights in flat disk shape attached to the ends of the bar. An adapter is provided at each end of the bar, such that weights can be removed and replaced. Another type of versatile weight equipment is a fixed stack machine. A weight stack with a hole in the middle of each piece is provided, with a stick with holes on it inserted into the stack at the holes. Each piece of weight has another hole horizontally. A user uses another stick to plug into the horizontal hole, thereby isolating the rest of the weight stack from being used.

[0005] The above weight equipment has a common problem, which is the lack of versatility among various equipment. Weights cannot be switched from the equipment of the first example to the equipment of the second example above. Weights designed for use with one type of equipment or machine do not work with others. Consequently, gyms are littered with different weight types. At-home or smaller gyms struggle to buy enough equipment to satisfy different exercise demands.

[0006] This invention aims to solve the above need and other needs in the art.

Abbreviations

[0007] cm: centimeter

[0008] CNC: computer numerical control

SUMMARY

[0009] The present invention provides exercise weights to be attached to different types of handles for use in exercising. The weight has a male part with a connecting mechanism to connect with a connecting mechanism on a hollow space in a handle, and optionally a hollow space with a connecting mechanism to connect with a connecting mechanism on a male part of another weight. Handles compatible with the weights can be a bar with at least a head with a

hollow space, a dumbbell with each weight on the side having a hollow space, or a kettle weight with a hollow space in the body.

[0010] In frequently provided embodiments, there is provided an exercise weight, comprising:

[0011] a body having a male part protruding from the body on one side of the body and a hollow space on the other side of the body,

[0012] wherein the male part has a connecting mechanism and the hollow space has a corresponding connecting mechanism,

[0013] wherein the male part is configured to connect with a corresponding hollow space on another weight,

[0014] wherein the hollow space is configured to receive a male part on another weight by connecting with the male part using the connecting mechanism.

[0015] In often provided embodiments, there is provided an exercise weight as above, wherein the body is of substantially cylindrical shape.

[0016] In often provided embodiments, there is provided an exercise weight as above, wherein the connecting mechanism is threads, retractable pins and holes, a snap fit, or an expanded ring with a corresponding expanded ring.

[0017] In frequently provided embodiments, there is provided an exercise weight as above, wherein the weight is provided in different masses, and the male parts and hollow spaces on each weight are configured to be of the same sizes and can be connected to each other.

[0018] In often provided embodiments, there is provided an exercise weight as above, further comprising knurling on the weight body.

[0019] In frequently provided embodiments, there is provided an exercise weight as above, further comprising a handle configured to attach to at least one weight by connecting with a male part of the weight.

[0020] In often provided embodiments, there is provided an exercise weight as above, wherein the handle is configured to attach to two weights.

[0021] In frequently provided embodiments, there is provided an exercise weight as above, wherein the handle comprises a dumbbell handlebar.

[0022] In often provided embodiments, there is provided an exercise weight as above, further comprising knurling on the dumbbell handlebar.

[0023] In frequently provided embodiments, there is provided an exercise weight as above, wherein the handle comprises a kettle weight handlebar.

[0024] In often provided embodiments, there is provided an exercise weight as above, further comprising knurling on the kettle weight handlebar.

[0025] In frequently provided embodiments, there is provided an exercise weight as above, wherein the handle comprises a mace bell handlebar.

[0026] In often provided embodiments, there is provided an exercise weight as above, further comprising knurling on the mace bell handlebar.

[0027] In frequently provided embodiments, there is provided an exercise weight, comprising:

[0028] a body having a male part protruding from the body on one side of the body,

[0029] wherein the male part has a connecting mechanism,

[0030] wherein the male part is configured to connect with a corresponding hollow space on another weight,

the corresponding hollow space has a connecting mechanism configured to connect with the connecting mechanism on the male part.

[0031] In often provided embodiments, there is provided an exercise weight as above, wherein the connecting mechanism is threads, retractable pins and holes, a snap fit, or an expanded ring with a corresponding expanded ring.

[0032] In frequently provided embodiments, there is provided an exercise weight as above, wherein the body is of substantially cylindrical shape.

[0033] In often provided embodiments, there is provided an exercise weight as above, further comprising knurling on the body.

BRIEF DESCRIPTION OF THE DRAWINGS

[0034] FIG. 1 is a top perspective view of weights of varying sizes according to embodiments herein.

[0035] FIG. 2 is a bottom perspective view of weights of varying sizes according to embodiments herein.

[0036] FIG. 3 is an exploded view of a dumbbell compatible for use with the weights in FIGS. 1 and 2.

[0037] FIG. 4 is an exploded view of a kettle weight compatible for use with the weights in FIGS. 1 and 2.

[0038] FIG. 5 is an exploded view of a mace compatible for use with the weights in FIGS. 1 and 2.

[0039] FIG. 6 is a side, exploded, closeup view of the head of the mace in FIG. 5.

[0040] FIG. 7 is a perspective, closeup view of the head of the mace in FIG. 5, without the weights.

DETAILED DESCRIPTION OF CERTAIN INVENTIVE EMBODIMENTS

[0041] This present invention is capable of being embodied in various forms. The description below of several embodiments is made with the understanding that the present disclosure is to be considered as an exemplification of the claimed subject matter and is not intended to limit the attached claims to the specific embodiments illustrated. The headings used throughout this disclosure are provided for convenience only and are not to be construed to limit the claims in any way. Embodiments illustrated under any heading may be combined with embodiments illustrated under any other heading.

[0042] As used herein, the verb “to comprise” in this description, claims, and other conjugations are used in its non-limiting sense to mean those items following the word are included, but items not specifically mentioned are not excluded.

[0043] Reference to an element by the indefinite article “a” or “an” does not exclude the possibility that more than one of the elements are present, unless the context clearly requires that there is one and only one of the elements. The indefinite article “a” or “an” thus usually means “at least one.” Additionally, the words “a” and “an” when used in the present document in concert with the words “comprising” or “containing” denote “one or more”.

[0044] As used herein in the specification and claims, including as used in the examples and unless otherwise expressly specified, all numbers may be read as if by prefaced by the word “about” or “approximately”, even if the term does not expressly appear. The phrase “about” or “approximately” may be used when describing magnitude

and/or position to indicate that the value and/or position described is within a reasonably expected range of values and/or positions.

[0045] As used herein, the phrase “male part” refers to a part protruding out of a body on a weight used in exercise.

[0046] All dimensions given herein are for illustrative purposes only and in no way will limit the inventions by these dimensions. It is to be understood that the invention may be constructed to have different dimensions than those provided herein and is still within the scope of the embodiments described herein. Drawings are not necessarily drawn to scale.

[0047] As used herein, like numerals indicate like components even though the components may be used in different manners or at different places. Where there are multiple components of the same nature, a numeral refers to one, some, or all of the components of the same nature, depending on the context.

[0048] All dimensions specified in this specification are by way of example only and not intended to be limiting. The actual size of the weights, handles, and other parts can vary to meet specific needs of various users.

[0049] In weight training, adding and reducing the number of weights on the equipment is a part of any exercise regime. For exercise equipment catered to more than one user, this is even more so. Typically, many exercise equipment are provided with the ability to exchange weights in and out. Locking mechanisms are provided to secure newly added weights.

[0050] Embodiments of this application relate to exercise weights with parts for assembling into different handles or into other weights of the same type. Assembly is by threads on male and female mating parts, by retractable pins and holes on corresponding mating parts, by a snap fit, or an expanded ring in place of threads on the inside of the hollow space and a corresponding ring on the male part of a weight. Other connecting mechanisms are contemplated. Each weight serves as a weight and as a locking mechanism by itself. In the drawings, for simplicity, threads are depicted as the connecting mechanism.

[0051] FIG. 1 is a perspective view of three (3) weights according to embodiments herein. The weight **100** has a body **105**, shown as substantially round and disk like, even though other suitable shapes are contemplated. On top of the weight **100** is a male part **102** having threads thereon. The male part **102** is fabricated and configured to mate with a hollow space having corresponding threads thereon. The male part **102** is of cylindrical shape and protrudes out of the body of the weight **100**. In place of thread, other connecting mechanisms such as retractable pins and holes, a snap fit, or an expanded ring with corresponding expanded ring may be used.

[0052] FIG. 2 is a bottom perspective view of the same three weights shown in FIG. 1. At the bottom of the weight **100** is a hollow space **103** having threads therein to mate with threads on a male part of another weight. The hollow space **103** is configured such that, when the male part **102** on another weight is inserted into the hollow space **103** and threaded with the threads on the hollow space, it will fit with the weight it inserts into to form an even weight stack.

[0053] On the outside of the weight body **101** is knurling **104**. Knurling **104** is a rough surface to give a texture to the exterior surface of the body **101**. When the weight **101** is being handled or turned to thread into place, knurling **104**

gives the necessary friction for easy handling. In FIGS. 1 and 2, knurling is shown as covering the bottom half of the weight outside surface, but knurling can also cover more or less surface of the weight. Preferably there is enough surface covered with knurling for a human hand's grip to facilitate attaching and removing the weights.

[0054] In an alternative embodiment, the weight does not include a hollow space at the bottom of the weight body. This embodiment allows the weight to attach to another weight or to a handle, but the weight itself will not have another weight attached to it.

[0055] In embodiments, weights of different sizes and masses are provided for use. The body of the weights may vary depending on the weight. The male part and the hollow space on each weight are configured to be of the same size and the same configuration, such that each male part can mate with each hollow space. Where there are multiple weights of varying sizes and shapes or the same size and shape, the weights can still be attached to each other due to the same male part and hollow space. The weights, upon attaching to each other by threading the male part with the hollow space, form a daisy chain of weight, which can be attached to a handle.

[0056] Weights according to embodiments herein are configured to attach to different handles for use in weightlifting exercise. Each weight has a male and a female connecting part capable of connecting to one another and/or to a suitable connecting part of a handle. The handle can be in various shapes, such that when the weights are attached, the resulting weight is a dumbbell, a kettle weight, or a mace, among other configurations.

[0057] FIG. 3 is an exploded view of weights attached to a handle, which is a dumbbell bar in this case. The connecting mechanism shown is threads. The dumbbell bar 200 has two small weights 201 attached to each end. The dumbbell bar 200 is a tube-like bar with a hollow space at each end of the bar shaped and sized to mate with the male part on each small weight 201. Each small weight 201 has a hollow space 203 at the bottom of the weight body, which mates with the male part 204 on the big weights 202. Small weights 201 threads into the dumbbell bar 200 while big weights 202 threads into the hollow space 203 on small weights 201. This process can continue with increasingly larger weights until the desired weight is reached. As shown in FIG. 3, the smaller weights 201 are attached to the handlebar 200 first, then larger weights 202 are attached to the smaller weights 201, even though this can be reverse.

[0058] FIG. 4 is an exploded view of an embodiment of the exercise weight as described herein attached to a kettle weight bar to form a kettle weight 300. A kettle weight bar 301 is provided with a kettle weight body 302 having a hollow space 303 with threads configured to mate with threads on a male part 306 of a first weight 304, the first weight 304 comprises a body 307 from which the male part 306 protrudes. A second weight 305 with a male part 306 mates with the hollow space 303 on the first weight 304 using threads on the male part and the hollow space. As shown in FIG. 4, the first weight 304 is larger than the second weight 305. However, a reverse order, i.e. the smaller weight attaching to the kettle weight handle and the larger weight attaching to the smaller weight, can also be used. This is true for all handle types. Upon assembly of at least one weight into the kettle weight bar, a complete kettle weight 300 is formed with the capability to add more

weights. Weights to be used with a kettle weight handle also have knurling on them for ease of grip by a human hand.

[0059] FIG. 5 is an exploded view of a mace having weights according to embodiments herein. Mace 400 has a long bar 404 with a mace head 403. The long bar 404 is shaped similar to a stick, with one end expanding into a head 403 with a funnel shape 406 attached to a cylindrical bottom 407. The funnel shape 406 acts as a transition area to give the mace 400 an aesthetic appearance. The mace head 403 is configured to mate with a male part 411 of a first weight 408 via a hollow space (not shown). A second weight 409 attaches to the first weight 408 by a male part 411 threading into the hollow space on the first weight. This can continue with more weights until the desired weight of the mace has been reached.

[0060] As shown in FIG. 5, second weight 409 does not have a hollow space at the bottom. This is an alternative embodiment of the weight. The weight according to this embodiment can be used as the last weight to be threaded onto the series of weights. A flat bottom without a hollow space gives a better-balanced surface on which the entire chain of weights and the handle can rest.

[0061] The mace 400 is characterized by a long bar handle. A person may hold the bar on the mace to move the mace in certain exercise routines. The mace bar 404 has an area having knurling 401 on it to facilitate gripping by hand. As shown in FIG. 5, knurling 401 covers the side of the mace distal from the mace head 403 spanning about 40% of the area of the bar. Knurling can cover the long bar for any amount of area and is strategically placed to facilitate different exercise routine. A bar covered completely in knurling can be gripped at both ends of the bar. A bar with knurling as shown in FIG. 5 can be gripped at the distal end of the bar.

[0062] At the end of the mace bar 404 is optionally a plug 410 fitted into a hollow space present at the end of the mace bar 404. In some embodiments, the mace bar 404 is hollow or has a small tunnel running along the inside of the bar from one end to another. The plug 410 keeps moisture and/or other objects or dust from entering the hollow tunnel inside the mace bar 404. Plug 410 may fit into the top of the mace bar 404 by pressing into the hollow space. Alternatively, plug 410 can fit into the top of the mace bar 404 by screwing into threads, or by welding, or by other fastening mechanisms. Plug 404 may be made of plastic, metal, alloy, or other suitable materials. If plug is made of metal or alloy, it should be made from the same materials with the mace and the weights, such that corrosion due to galvanic reaction is avoided. It is noted that the plug is optional, and in some embodiments the mace bar can be solid.

[0063] FIG. 6 is a side exploded view of the mace head and two weights attached to the mace head in a daisy chain configuration. The hollow space 402 in the mace head has threads configured to mate with threads on the male part 411 of a first weight 408. When the first weight 408 has been screwed into the hollow space 402, the first weight 408 and the mace head 403 form a mace with weight ready for use. However, as shown in FIG. 6, additional weights can be attached to the mace by mating the male part on a different weight with the hollow space on the first weight 408. Second weight 409, as shown herein is an embodiment of the weight without a hollow space, is screwed into the hollow space (not shown) on the first weight 408, forming a mace with two weights attached to the mace head 403. More weights

can be attached to the mace long bar using weights with a male part and a hollow space, with the male part on one weight screwed into the hollow space of another weight. The process can continue until the last weight with a male part but without a hollow space is attached to a weight already on the mace. It is noted that the last weight to be added into the mace bar does not have to be a weight without a hollow space.

[0064] Different connecting mechanisms may be used for connecting the weights. For example, retractable pins may be present on the male part of a weight while corresponding holes are present on the hollow space of another weight. A user pushes the weight with the male part into the hollow space of the corresponding weight and line up the pins with the holes. The pins are pushed in while the male part enter the hollow space, and expands out to fill the holes, thereby locking the weights in place. Other connecting mechanisms may also be used, including snap fit, or an expanded ring in place of threads on the inside of the hollow space and a corresponding ring on the male part of a weight.

[0065] Apart from dumbbell handlebar, mace handlebar and kettle weight handlebar, other handlebars can be configured to be used with the weights according to embodiments herein. For example, the handlebar can mimic a sledgehammer, a barbell, or other types of handles.

[0066] Weights are made to have different weights and different sizes to meet the demand of each user. Handles are made to correspond to weights. By way of example only, weights are made available at 3-, 5-, and 10-pound pieces. Dumbbell handlebar weighs, for example, 6 pounds. The kettle weight handlebar weighs, for example, 8.5 pounds. A small mace handlebar weighs 6 pounds and is 25 inches long. A large mace handlebar weighs 10 pounds and is 35 inches long. These values and dimensions are for example only and shall in no way limit the invention herein.

[0067] Parts are manufactured with computer numerical control (CNC) machining, tube benders and welder power supply. All parts are chrome coated to avoid rust and abrasion. Knurling on weights and knurling on handlebars are different to facilitate different grip demand. The force required to turn the weights is different than the force exerted onto handlebars during exercise, therefore the knurling is configured at each location to fit the need.

[0068] Weights together with handles as described in embodiments herein can be used as exercise tools, including strength training, muscle developments, function development, or fat burning. A variety of exercise routines can be performed with the weights attached to the various handles compatible with the weights.

[0069] Weights and handles according to embodiments herein can be disassembled for convenient transportation and assembled together when used. All weights are suitable for use with all handles and thus the number of parts required to meet certain weight needs is reduced. A user and/or a gym requires less parts to provide the same weight variety needed, thus greatly reducing material expenses and labor costs.

[0070] In a first embodiment, an exercise weight is provided, the exercise weight comprises:

[0071] a body having a male part protruding from the body on one side of the body and a hollow space on the other side of the body,

[0072] wherein the male part has a connecting mechanism and the hollow space has a corresponding connecting mechanism,

[0073] wherein the male part is configured to connect with a corresponding hollow space on another weight,

[0074] wherein the hollow space is configured to receive a male part on another weight by connecting with the male part using the connecting mechanism.

[0075] In a second embodiment, the first embodiment includes the body is of substantially cylindrical shape.

[0076] In a third embodiment, the first embodiment includes the connecting mechanism is threads, retractable pins and holes, a snap fit, or an expanded ring with a corresponding expanded ring.

[0077] In a fourth embodiment, the first embodiment includes the weight is provided in different masses, and the male parts and hollow spaces on each weight are configured to be of the same sizes and can be connected to each other.

[0078] In a fifth embodiment, the first embodiment further comprises knurling on the weight body.

[0079] In a sixth embodiment, the first embodiment further comprises a handle configured to attach to at least one weight by connecting with a male part of the weight.

[0080] In a seventh embodiment, the sixth embodiment includes the handle is configured to attach to two weights.

[0081] In an eighth embodiment, the seventh embodiment includes the handle comprises a dumbbell handlebar.

[0082] In a ninth embodiment, the eighth embodiment further comprises knurling on the dumbbell handlebar.

[0083] In a tenth embodiment, the sixth embodiment includes the handle comprises a kettle weight handlebar.

[0084] In an eleventh embodiment, the tenth embodiment further comprises knurling on the kettle weight handlebar.

[0085] In a twelfth embodiment, the sixth embodiment includes the handle comprises a mace bell handlebar.

[0086] In a thirteenth embodiment, the twelfth embodiment further comprises knurling on the mace bell handlebar.

[0087] In a fourteenth embodiment, there is provided an exercise weight, comprising:

[0088] a body having a male part protruding from the body on one side of the body,

[0089] wherein the male part has a connecting mechanism,

[0090] wherein the male part is configured to connect with a corresponding hollow space on another weight, the corresponding hollow space has a connecting mechanism configured to connect with the connecting mechanism on the male part.

[0091] In a fifteenth embodiment, the fourteenth embodiment includes the connecting mechanism is threads, retractable pins and holes, a snap fit, or an expanded ring with a corresponding expanded ring.

[0092] In a sixteenth embodiment, the fourteenth embodiment includes the body is of substantially cylindrical shape.

[0093] In a seventeenth embodiment, the fourteenth embodiment further comprises knurling on the body.

[0094] While the present invention has been discussed in detail with reference to certain embodiments, other embodiments are possible. Therefore, the scope of the appended claims should not be limited to the description of the preferred embodiments contained in this disclosure.

[0095] All references, including publications, patent applications, and patents cited herein are hereby incorporated by reference to the same extent as if each reference were

individually and specifically indicated to be incorporated by reference and were set forth in its entirety herein.

[0096] It will be readily apparent to those skilled in the art that a number of modifications and changes may be made without departing from the spirit and the scope of the present invention. It is to be understood that any ranges, ratios, and range of ratios that can be derived from any of the data disclosed herein represent further embodiments of the present disclosure and are included as part of the disclosure as though they were explicitly set forth. This includes ranges that can be formed that do or do not include a finite upper and/or lower boundary. Accordingly, a person of ordinary skill in the art will appreciate that such values are unambiguously derivative from the data presented herein.

What is claimed is:

1. An exercise weight comprising:
a body having a male part protruding from the body on one side of the body and a hollow space on the other side of the body,
wherein the male part has a connecting mechanism and the hollow space has a corresponding connecting mechanism,
wherein the male part is configured to connect with a corresponding hollow space on another weight,
wherein the hollow space is configured to receive a male part on another weight by connecting with the male part using the connecting mechanism.
2. The exercise weight of claim 1, wherein the body is of substantially cylindrical shape.
3. The exercise weight of claim 1, wherein the connecting mechanism is threads, retractable pins and holes, a snap fit, or an expanded ring with a corresponding expanded ring.
4. The exercise weight of claim 1, wherein the weight is provided in different masses, and the male parts and hollow spaces on each weight are configured to be of the same sizes and can be connected to each other.

5. The exercise weight of claim 1, further comprising knurling on the weight body.

6. The exercise weight of claim 1, further comprising a handle configured to attach to at least one weight by connecting with a male part of the weight.

7. The exercise weight of claim 6, wherein the handle is configured to attach to two weights.

8. The exercise weight of claim 7, wherein the handle comprises a dumbbell handlebar.

9. The exercise weight of claim 8, further comprising knurling on the dumbbell handlebar.

10. The exercise weight of claim 6, wherein the handle comprises a kettle weight handlebar.

11. The exercise weight of claim 10, further comprising knurling on the kettle weight handlebar.

12. The exercise weight of claim 6, wherein the handle comprises a mace bell handlebar.

13. The exercise weight of claim 12, further comprising knurling on the mace bell handlebar.

14. An exercise weight, comprising:

a body having a male part protruding from the body on one side of the body,

wherein the male part has a connecting mechanism, wherein the male part is configured to connect with a corresponding hollow space on another weight, the corresponding hollow space has a connecting mechanism configured to connect with the connecting mechanism on the male part.

15. The exercise weight of claim 14, wherein the connecting mechanism is threads, retractable pins and holes, a snap fit, or an expanded ring with a corresponding expanded ring.

16. The exercise weight of claim 14, wherein the body is of substantially cylindrical shape.

17. The exercise weight of claim 14, further comprising knurling on the body.

* * * * *